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State Diagram:

AI Implementation Approach: (i can finish this later)

Our guard AI was implemented using a state machine in tandem with Unity’s NavMesh navigation system. The agent goes through four states: Roaming, Suspicious, Confused, and Alerted.

The agent uses a detection system consisting of a vision cone in front of it, in conjunction with a small circular radius around it. While Roaming, guards will patrol between predefined waypoints using NavMesh pathfinding. If the player enters the detection area and is within line of sight, the guard enters the Suspicious state. While Suspicious, the guard will remain stationary and track the player’s movement, while the suspicion level slowly increases. If suspicion reaches the maximum threshold, the guard enters the Alerted state and actively pursues the player in a near inescapable manner, essentially being a loss condition. However, if the player were to briefly enter guards vision, triggering the Suspicious state, but then escapes before they become Alerted, they get put into the Confusion state. When Confused, the agent will investigate by traveling to the player’s last known location, promptly returning to their Roaming state after a brief delay. Visual feedback was incorporated through state-based color changes, and the suspicion levels are indicated by the opacity of the agents’ vision indicators.

Reflections:

Bryce Han:

My contributions were creating the agent checkpoints for pathing purposes, along with configuring the checkpoints and AI agents to navigate between them. The most interesting thing I learned was how Unity’s NavMesh system handles pathfinding and how small adjustments to waypoint placement and agent settings can significantly affect movement behavior and reliability.

Shawn Kramer:

My contributions were the map creation and the win and lose conditions, and everything tied to them. The most interesting thing I learned was about rebaking the NavMesh while creating the map to make sure it was navigable. I also learned while doing some things for the first time, such as resetting the scene and pausing the scene for the win screen.

Ivan Li:

My contribution to the project was primarily creation of the agent states/behaviors and all of the vision implementation. The most interesting thing I learned from this assignment was regarding the implementation of vision, specifically the visualization of it. My approach to the vision might not have been the best, but I figured it would be easiest to handle it all via script. The state machine was not all that unfamiliar to me, as I've worked with state machines in some other projects, but it was a bit tricky to set it all up myself.

